Name: Ekta Narayan Sawant
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Subject: ASD&D

Experiment 02

Aim: To containerise the application using docker

This experiment maps to LO4.

LO4: Containerize, orchestrate, and deploy applications using Docker, Docker Compose, Kubernetes, and cloud infrastructure tools.

Theory:

Docker is an open-source containerization platform that enables applications to run in isolated environments called containers. A container includes the application along with all its required libraries, dependencies, and configuration files, ensuring consistent execution across different systems. Since containers share the host operating system's kernel, they are lightweight, portable, and start much faster than traditional virtual machines.

A Docker image is a read-only template that contains everything needed to create a container, including the operating system, software packages, and application files. Images are stored in repositories such as Docker Hub and can be downloaded using the docker pull command. Once an image is available locally, it can be used to create one or more containers.

Docker provides several commands for managing images and containers. The docker --version command is used to verify the installed Docker version, while docker images display all locally available Docker images. The docker search command allows users to search Docker Hub for official or community-maintained images, and docker pull downloads the required image from the repository.

Containers are created and started using the docker run command. Running a container interactively (docker run -it ubuntu) opens a Linux shell inside the container, allowing users to execute Linux commands as if they were working on a standalone Ubuntu system. Commands such as pwd display the current working directory, ls lists the files and directories, mkdir creates a new directory, cd changes the current directory, touch creates an empty file, echo writes text into a file, and cat displays the contents of the file.

These basic Docker and Linux commands help users understand how containers work, how to interact with a running container, and how to perform file and directory operations within an isolated environment. This forms the foundation for deploying, testing, and managing applications using Docker in real-world DevOps workflows.

Docker Installation Steps

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1. Check system requirements: Ensure your operating system supports Docker and virtualization.
2. Enable virtualization (Windows): Turn on Intel VT-x or AMD-V in the BIOS/UEFI settings.
3. Install WSL 2 (Windows only): WSL 2 provides the Linux environment required by Docker Desktop.
4. Download and install Docker Desktop: Install Docker Desktop from the official Docker website and follow the setup instructions.
5. Restart your computer: Restart if prompted to complete the installation.
6. Launch Docker Desktop: Open Docker Desktop and wait for the Docker Engine to start.

What is a docker image?

A Docker image is a read-only template used to create Docker containers. It contains everything required to run an application, including the operating system, application code, libraries, dependencies, configuration files, and runtime environment. Docker images are stored in repositories such as Docker Hub and can be downloaded using the docker pull command. Since images are portable and platform-independent, they ensure that applications run consistently across different environments.

A Docker image itself cannot execute applications. When the docker run command is used, Docker creates a container, which is a running instance of the image. Multiple containers can be created from the same image, making Docker an efficient solution for application deployment, testing, and scaling. Images are also built in layers, which reduces storage requirements and speeds up image downloads and updates by reusing common layers.

What is containerization?

Containerization is a virtualization technology that packages an application along with all its dependencies, libraries, configuration files, and runtime environment into a single, isolated unit called a container. This ensures that the application runs consistently across different environments, regardless of the underlying operating system or hardware.

Unlike traditional virtual machines, containers share the host operating system's kernel, making them lightweight, fast, and resource-efficient. Containers start within seconds, require less memory, and allow multiple applications to run independently on the same system without conflicts. Docker is one of the most widely used platforms for containerization, enabling developers to build, deploy,

and manage applications efficiently. Containerization improves application portability, scalability, and reliability, making it an essential technology in modern software development and DevOps practices.

What is the difference between docker image and docker container?

Docker Image	Docker Container
A read-only template used to create containers.	A running instance of a Docker image.
Contains application code, libraries, dependencies, and configuration files.	Contains the running application along with its processes and data.
Cannot execute applications by itself.	Executes the application inside an isolated environment.
Created using a Dockerfile.	Created using the docker run command.
Stored in image repositories such as Docker Hub.	Exists only while it is created on the host system.
Multiple containers can be created from a single image.	Each container is an independent instance of the image.
Read-only and cannot be modified directly.	Read-write; changes can be made while the container is running.
Example: ubuntu, nginx, python, mysql images.	Example: A running Ubuntu container created using docker run -it ubuntu.

Implementation:

1)check docker version

```
C:\Users\15L>docker --version
Docker version 29.6.1, build 8900f1d
```

2)images

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```
C:\Users\15L>docker images

IMAGE                                ID             DISK USAGE   CONTENT SIZE   EXTRA
alpine:latest                        d529dd0c6e55       8.42MB             0B
aryan077800/mysql:latest             4babd5755458        941MB             0B
aryan077800/sample-flask-app:v1      657cf5304039        141MB             0B    U
dockercommands-backend:latest        fcd302b5399c       1.11GB             0B    U
dockercommands-frontend:latest       054cf831e468       62.4MB             0B    U
hello-world:latest                   e2ac70e7319a       10.1kB             0B    U
httpd:latest                         4613a77dcb46        117MB             0B    U
mysql:8.0                            6cd09145362d        799MB             0B    U
mysql:latest                         4babd5755458        941MB             0B
nginx:1.0                            fcbd66995946        169MB             0B
nginx:latest                         4e5db4761e0f        161MB             0B    U
node:latest                          c3978d05bc68       1.1GB              0B    U
phpmyadmin/phpmyadmin:latest         e66b1f5a8c58        742MB             0B    U
redis:latest                         3848bb827e71        140MB             0B    U
sample-flask-app:v1                  657cf5304039        141MB             0B    U
ubuntu:latest                        de7345b16e94        100MB             0B    U
```

3)

```
C:\Users\15L>docker pull ubuntu
Using default tag: latest
latest: Pulling from library/ubuntu
Digest: sha256:3131b4cc82a783df6c9df078f86e01819a13594b865c2cad47bd1bca2b7063bb
Status: Image is up to date for ubuntu:latest
docker.io/library/ubuntu:latest

What's next:
    View a summary of image vulnerabilities and recommendations → docker scout quickview ubuntu
```

4)

```
C:\Users\15L>docker run -it ubuntu
root@bdb8d2011273:/# pwd
/
root@bdb8d2011273:/# ls
bin  boot  dev  etc  home  lib  lib64  media  mnt  opt  proc  root  run  sbin  srv  sys  tmp  usr  var
```

```
root@a6b78cebe09c:/home/T21_86# exit
exit
```

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```
root@bdb8d2011273:/# cd home
root@bdb8d2011273:/home# mkdir T22_103
root@bdb8d2011273:/home# cd T22_103
root@bdb8d2011273:/home/T22_103# touch roll_no_103
root@bdb8d2011273:/home/T22_103# echo "This is my text" >roll_no_103
root@bdb8d2011273:/home/T22_103# cat roll_no_103.txt
cat: roll_no_103.txt: No such file or directory
root@bdb8d2011273:/home/T22_103# cat roll_no_103
This is my text
root@bdb8d2011273:/home/T22_103# exit
exit
```

```
C:\Users\15L>docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
```

```
C:\Users\15L>docker ps -a
CONTAINER ID   IMAGE                    COMMAND                  CREATED         STATUS                          PORTS                                 NAMES
bdb8d2011273   ubuntu                   "/bin/bash"              5 minutes ago   Exited (0) About a minute ago                                         compassionate_ride
7f71697c02a2   ubuntu                   "/bin/bash"              2 hours ago     Exited (0) 2 hours ago                                                silly_burnell
cc7f4a16e717   ubuntu                   "/bin/bash"              2 hours ago     Exited (0) 2 hours ago                                                blissful_kapitsa
56997ea323e9   ubuntu                   "/bin/bash"              2 hours ago     Exited (0) 2 hours ago                                                beautiful_mclean
9dff908e1dda   dockercommands-frontend  "/docker-entrypoint.…"   23 hours ago    Exited (255) 2 hours ago        0.0.0.0:3000->80/tcp                  frontend
c5e99386e34e   dockercommands-backend   "docker-entrypoint.s…"   23 hours ago    Exited (255) 2 hours ago        0.0.0.0:5000->5000/tcp                backend
44ac4dab94c8   phpmyadmin/phpmyadmin    "/docker-entrypoint.…"   2 days ago      Exited (255) 24 hours ago       0.0.0.0:8083->80/tcp                  phpmyadmin
62d89ff8623b   nginx:latest             "/docker-entrypoint.…"   2 days ago      Exited (255) 24 hours ago       0.0.0.0:8082->80/tcp                  webserver
cdad494367a5   mysql:8.0                "docker-entrypoint.s…"   2 days ago      Exited (255) 24 hours ago       33060/tcp, 0.0.0.0:3307->3306/tcp     mysqlserver
5bee38c410df   redis:latest             "docker-entrypoint.s…"   2 days ago      Exited (255) 24 hours ago       0.0.0.0:6380->6379/tcp                redisserver
1c1a8f2165ff   mysql:8.0                "docker-entrypoint.s…"   2 days ago      Exited (255) 2 days ago         33060/tcp, 0.0.0.0:3307->3306/tcp     compose-demo-database-1
7b74d7f4d81f   nginx                    "/docker-entrypoint.…"   8 days ago      Exited (255) 2 days ago         80/tcp                                bold_poincare
9643d3b82fbd   hello-world              "/hello"                 8 days ago      Exited (0) 8 days ago                                                 wizardly_wiles
c7421e030536   ubuntu                   "/bin/bash"              8 days ago      Exited (255) 2 days ago                                               myubuntu
10102c1f33de   ubuntu                   "bin/bash"               9 days ago      Exited (0) 9 days ago                                                 great_chebyshev
94a929572579   ubuntu                   "/bin/bash"              9 days ago      Exited (0) 9 days ago                                                 gifted_roentgen
2f937354l8dbe  redis                    "docker-entrypoint.s…"   10 days ago     Exited (255) 9 days ago         6379/tcp                              sad_cartwright
885d5e85d231   httpd                    "httpd-foreground"       10 days ago     Created                                                               reverent_snyder
452bcec1feb4   sample-flask-app:v1      "flask run --host=0.…"   10 days ago     Exited (0) 10 days ago                                                my_new_app
c765c3d83e84   ubuntu                   "/bin/bash"              10 days ago     Exited (255) 9 days ago                                               hungry_blackburn
b371b47ddd31   ubuntu                   "/bin/bash"              10 days ago     Exited (255) 9 days ago                                               distracted_clarke
5ba8b97fc7a2   ubuntu                   "\\bin\\bash"            10 days ago     Created                                                               focused_wilson
1c355d58581d   sample-flask-app:v1      "flask run --host=0.…"   4 months ago    Exited (137) 10 days ago                                              stoic_jang
498c2de19d41   redis                    "docker-entrypoint.s…"   4 months ago    Exited (255) 10 days ago        6379/tcp                              vibrant_goldberg
e81c8ac65cd6   httpd                    "httpd-foreground"       8 months ago    Exited (0) 8 months ago                                               jolly_herschel
f90bf8584ef0   httpd                    "httpd-foreground"       8 months ago    Exited (0) 8 months ago                                               inspiring_montalcini
c1558a5bdfc7   httpd                    "httpd-foreground"       8 months ago    Exited (0) 8 months ago                                               compassionate_cerf
61c302b4db0a   httpd                    "httpd-foreground"       8 months ago    Exited (0) 8 months ago                                               suspicious_heisenberg
a916b1bc1d13   httpd                    "httpd-foreground"       8 months ago    Exited (0) 8 months ago                                               nice_roentgen
004bcde700ff   httpd                    "httpd-foreground"       8 months ago    Exited (0) 8 months ago                                               sad_noyce
c8cb01650812   httpd                    "httpd-foreground"       8 months ago    Exited (0) 8 months ago                                               flamboyant_pike
d60547f1917e   httpd                    "httpd-foreground"       8 months ago    Exited (0) 8 months ago                                               intelligent_greider
05e923e0452c   httpd                    "httpd-foreground"       8 months ago    Exited (0) 8 months ago                                               zealous_bouman
250bb0ef6efe   httpd                    "httpd-foreground"       8 months ago    Exited (0) 8 months ago                                               sad_elion
74b68ec74672   httpd                    "httpd-foreground"       8 months ago    Exited (0) 8 months ago                                               hungry_jang
55a80be3d445   httpd                    "httpd-foreground"       8 months ago    Exited (0) 8 months ago                                               youthful_matsumoto
4ed8cfaded6b   httpd                    "httpd-foreground"       8 months ago    Exited (0) 8 months ago                                               sleepy_gates
2fa633478597   ca2b0f26964c             "/bin/bash"              2 years ago     Exited (255) 11 months ago                                            unruffled_wilson
ee37479b71cd   node                     "docker-entrypoint.s…"   2 years ago     Exited (0) 2 years ago                                                sarth
9c7ca042694c   node                     "docker-entrypoint.s…"   2 years ago     Exited (0) 2 years ago                                                lucid_faraday
f4b59b6cf1f4   d1397258b209             "docker-entrypoint.s…"   2 years ago     Exited (255) 11 months ago      6379/tcp                              musing_archimedes
d820ef6a1a06   d2c94e258dcb             "/hello"                 2 years ago     Exited (0) 2 years ago                                                bb71b1a51783
bb71b1a51783   d2c94e258dcb             "/hello"                 2 years ago     Exited (0) 2 years ago                                                silly_wozniak
922a7dec7144   d1397258b209             "docker-entrypoint.s…"   2 years ago     Exited (255) 2 days ago         6379/tcp                              festive_hertz
97abe06c14c7   d2c94e258dcb             "/hello"                 2 years ago     Exited (0) 8 days ago                                                 b9f4789574a5
ae743b95da80   d2c94e258dcb             "/hello"                 2 years ago     Exited (0) 2 years ago                                                mystifying_ramanujan
b9f4789574a5   d2c94e258dcb             "/hello"                 2 years ago     Exited (0) 2 years ago                                                interesting_yalow
```

```
C:\Users\15L>docker start silly_burnell
silly_burnell
```

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```
C:\Users\15L>docker inspect silly_burnell
[
    {
        "Id": "7f71697c02a277f99c0de7e414093ed594b3865db48a034c0622c46020fe136f",
        "Created": "2026-07-30T06:23:17.692077654Z",
        "Path": "/bin/bash",
        "Args": [],
        "State": {
            "Status": "running",
            "Running": true,
            "Paused": false,
            "Restarting": false,
            "OOMKilled": false,
            "Dead": false,
            "Pid": 602,
            "ExitCode": 0,
            "Error": "",
            "StartedAt": "2026-07-30T08:10:19.668553545Z",
            "FinishedAt": "2026-07-30T06:24:48.240884889Z"
        },
        "Image": "sha256:de7345b16e942e22044c6ba053020ec85ae879984860a9918517d54eb6cef851",
        "ResolvConfPath": "/var/lib/docker/containers/7f71697c02a277f99c0de7e414093ed594b3865db48a034c0622c46020fe136f/resolv.conf",
        "HostnamePath": "/var/lib/docker/containers/7f71697c02a277f99c0de7e414093ed594b3865db48a034c0622c46020fe136f/hostname",
        "HostsPath": "/var/lib/docker/containers/7f71697c02a277f99c0de7e414093ed594b3865db48a034c0622c46020fe136f/hosts",
        "LogPath": "/var/lib/docker/containers/7f71697c02a277f99c0de7e414093ed594b3865db48a034c0622c46020fe136f/7f71697c02a277f99c0de7e414093ed594b3865db48a034c0622c46020fe136f-json.log",
        "Name": "/silly_burnell",
        "RestartCount": 0,
        "Driver": "overlay2",
        "Platform": "linux",
        "MountLabel": "",
        "ProcessLabel": "",
        "AppArmorProfile": "",
        "ExecIDs": null,
        "HostConfig": {
            "Binds": null,
            "ContainerIDFile": "",
            "LogConfig": {
                "Type": "json-file",
                "Config": {}
            },
            "NetworkMode": "bridge",
            "PortBindings": {},
            "RestartPolicy": {
                "Name": "no",
                "MaximumRetryCount": 0
            },
            "AutoRemove": false,
```

```
                "PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin"
            ],
            "Cmd": [
                "/bin/bash"
            ],
            "Image": "ubuntu",
            "Volumes": null,
            "WorkingDir": "",
            "Entrypoint": null,
            "Labels": {
                "org.opencontainers.image.created": "2026-07-13T16:06:30.499069+00:00",
                "org.opencontainers.image.description": "The Ubuntu container image maintained by Canonical\n\nUbuntu is a Debian-based Linux operating system that runs from the desktop to the cloud, to all yo
ur internet connected things.\nIt is the world's most popular operating system across public clouds and OpenStack clouds.\nIt is the number one platform for containers; from Docker to Kubernetes to LXD, Ubuntu
 can run your containers at scale.\nFast, secure and simple, Ubuntu powers millions of PCs worldwide.\n",
                "org.opencontainers.image.title": "ubuntu",
                "org.opencontainers.image.version": "26.04"
            }
        },
        "NetworkSettings": {
            "SandboxID": "2dac9bb8744438d1228d43923f8bdbf7477d2410994ca991af6fbfca467e3610",
            "SandboxKey": "/var/run/docker/netns/2dac9bb87444",
            "Ports": {},
            "Networks": {
                "bridge": {
                    "IPAMConfig": null,
                    "Links": null,
                    "Aliases": null,
                    "DriverOpts": null,
                    "GwPriority": 0,
                    "NetworkID": "d47335ffa525912e17f926cd9263254a0768784b7b76bef869d78104ecd5b627",
                    "EndpointID": "4d31ee8d75d4c0b207cdc99a90ba315778c6fc2d7b0ba3641226ae5498a51dc4",
                    "Gateway": "172.17.0.1",
                    "IPAddress": "172.17.0.2",
                    "MacAddress": "72:2a:6d:54:db:e0",
                    "IPPrefixLen": 16,
                    "IPv6Gateway": "",
                    "GlobalIPv6Address": "",
                    "GlobalIPv6PrefixLen": 0,
                    "DNSNames": null
                }
            }
        }
    }
]
```

```
CONTAINER ID   NAME             CPU %     MEM USAGE / LIMIT   MEM %     NET I/O         BLOCK I/O     PIDS
a6b78cebe09c   gifted_shamir    0.00%     960KiB / 5.635GiB   0.02%     1.17kB / 126B   545kB / 0B    1
```

```
C:\Users\15L>docker rm -f silly_burnell
silly_burnell
```

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```
C:\Users\15L>docker image prune
WARNING! This will remove all dangling images.
Are you sure you want to continue? [y/N] n
```

```
C:\Users\15L>docker container ls -a
CONTAINER ID   IMAGE                      COMMAND                  CREATED          STATUS                      PORTS                                 NAMES
bdb8d2011273   ubuntu                     "/bin/bash"              14 minutes ago   Exited (0) 10 minutes ago                                         compassionate_ride
cc7f4a16e717   ubuntu                     "/bin/bash"              2 hours ago      Exited (0) 2 hours ago                                            blissful_kapitsa
56997ea323e9   ubuntu                     "/bin/bash"              2 hours ago      Exited (0) 2 hours ago                                            beautiful_mclean
9dff908e1dda   dockercommands-frontend    "/docker-entrypoint.…"   23 hours ago     Exited (255) 2 hours ago    0.0.0.0:3000->80/tcp                  frontend
c5e99386e34e   dockercommands-backend     "docker-entrypoint.s…"   23 hours ago     Exited (255) 2 hours ago    0.0.0.0:5000->5000/tcp                backend
44ac4dab94c8   phpmyadmin/phpmyadmin      "/docker-entrypoint.…"   2 days ago       Exited (255) 24 hours ago   0.0.0.0:8083->80/tcp                  phpmyadmin
62d89ff8623b   nginx:latest               "/docker-entrypoint.…"   2 days ago       Exited (255) 24 hours ago   0.0.0.0:8082->80/tcp                  webserver
cdad494367a5   mysql:8.0                  "docker-entrypoint.s…"   2 days ago       Exited (255) 24 hours ago   33060/tcp, 0.0.0.0:3307->3306/tcp     mysqlserver
5bee38c410df   redis:latest               "docker-entrypoint.s…"   2 days ago       Exited (255) 24 hours ago   0.0.0.0:6380->6379/tcp                redisserver
1c1a8f2165ff   mysql:8.0                  "docker-entrypoint.s…"   2 days ago       Exited (255) 2 days ago     33060/tcp, 0.0.0.0:3307->3306/tcp     compose-demo-database-1
7b74d7f4d81f   nginx                      "/docker-entrypoint.…"   8 days ago       Exited (255) 2 days ago     80/tcp                                bold_poincare
9643d3b82fbd   hello-world                "/hello"                 8 days ago       Exited (0) 8 days ago                                             wizardly_wiles
c7421e030536   ubuntu                     "/bin/bash"              8 days ago       Exited (255) 2 days ago                                           myubuntu
10102c1f33de   ubuntu                     "bin/bash"               9 days ago       Exited (0) 9 days ago                                             great_chebyshev
94a929572579   ubuntu                     "/bin/bash"              9 days ago       Exited (0) 9 days ago                                             gifted_roentgen
2f9373548dbe   redis                      "docker-entrypoint.s…"   10 days ago      Exited (255) 9 days ago     6379/tcp                              sad_cartwright
885d5e85d231   httpd                      "httpd-foreground"       10 days ago      Created                                                           reverent_snyder
452bcec1feb4   sample-flask-app:v1        "flask run --host=0.…"   10 days ago      Exited (0) 10 days ago                                            my_new_app
c765c3d83e84   ubuntu                     "/bin/bash"              10 days ago      Exited (255) 9 days ago                                           hungry_blackburn
b371b47ddd31   ubuntu                     "/bin/bash"              10 days ago      Exited (255) 9 days ago                                           distracted_clarke
5ba8b97fc7a2   ubuntu                     "\\bin\\bash"            10 days ago      Created                                                           focused_wilson
1c355d58581d   sample-flask-app:v1        "flask run --host=0.…"   4 months ago     Exited (137) 10 days ago                                          stoic_jang
498c2de19d41   redis                      "docker-entrypoint.s…"   4 months ago     Exited (255) 10 days ago    6379/tcp                              vibrant_goldberg
e81c8ac65cd6   httpd                      "httpd-foreground"       8 months ago     Exited (0) 8 months ago                                           jolly_herschel
f90bf8584ef0   httpd                      "httpd-foreground"       8 months ago     Exited (0) 8 months ago                                           inspiring_montalcini
c1558a5bdfc7   httpd                      "httpd-foreground"       8 months ago     Exited (0) 8 months ago                                           compassionate_cerf
61c302b4db0a   httpd                      "httpd-foreground"       8 months ago     Exited (0) 8 months ago                                           suspicious_heisenberg
a916b1bc1d13   httpd                      "httpd-foreground"       8 months ago     Exited (0) 8 months ago                                           nice_roentgen
004bcde700ff   httpd                      "httpd-foreground"       8 months ago     Exited (0) 8 months ago                                           sad_noyce
c8cb01650812   httpd                      "httpd-foreground"       8 months ago     Exited (0) 8 months ago                                           flamboyant_pike
d60547f1917e   httpd                      "httpd-foreground"       8 months ago     Exited (0) 8 months ago                                           intelligent_greider
05e923e0452c   httpd                      "httpd-foreground"       8 months ago     Exited (0) 8 months ago                                           zealous_bouman
250bb0ef6efe   httpd                      "httpd-foreground"       8 months ago     Exited (0) 8 months ago                                           sad_elion
74b68ec74672   httpd                      "httpd-foreground"       8 months ago     Exited (0) 8 months ago                                           hungry_jang
55a80be3d445   httpd                      "httpd-foreground"       8 months ago     Exited (0) 8 months ago                                           youthful_matsumoto
4ed8cfaded6b   httpd                      "httpd-foreground"       8 months ago     Exited (0) 8 months ago                                           sleepy_gates
2fa633478597   ca2b0f26964c               "/bin/bash"              2 years ago      Exited (255) 11 months ago                                        unruffled_wilson
ee37479b71cd   node                       "docker-entrypoint.s…"   2 years ago      Exited (0) 2 years ago                                            sarth
9c7ca042694c   node                       "docker-entrypoint.s…"   2 years ago      Exited (0) 2 years ago                                            lucid_faraday
f4b59b6cf1f4   d1397258b209               "docker-entrypoint.s…"   2 years ago      Exited (255) 11 months ago  6379/tcp                              musing_archimedes
d820ef6a1a06   d2c94e258dcb               "/hello"                 2 years ago      Exited (0) 2 years ago                                            bb71b1a51783
bb71b1a51783   d2c94e258dcb               "/hello"                 2 years ago      Exited (0) 2 years ago                                            silly_wozniak
922a7dec7144   d1397258b209               "docker-entrypoint.s…"   2 years ago      Exited (255) 2 days ago     6379/tcp                              festive_hertz
97abe06c14c7   d2c94e258dcb               "/hello"                 2 years ago      Exited (0) 8 days ago                                             b9f4789574a5
ae743b95da80   d2c94e258dcb               "/hello"                 2 years ago      Exited (0) 2 years ago                                            mystifying_ramanujan
b9f4789574a5   d2c94e258dcb               "/hello"                 2 years ago      Exited (0) 2 years ago                                            interesting_yalow
```

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Subject: ASD&D

```
C:\Users\15L>docker info
Client:
 Version:    29.6.1
 Context:    desktop-linux
 Debug Mode: false
 Plugins:
  agent: Docker AI Agent Runner (Docker Inc.)
    Version:  v1.98.0
    Path:     C:\Program Files\Docker\cli-plugins\docker-agent.exe
  ai: Docker AI Agent - Ask Gordon (Docker Inc.)
    Version:  v1.27.0
    Path:     C:\Program Files\Docker\cli-plugins\docker-ai.exe
  buildx: Docker Buildx (Docker Inc.)
    Version:  v0.35.0-desktop.2
    Path:     C:\Program Files\Docker\cli-plugins\docker-buildx.exe
  compose: Docker Compose (Docker Inc.)
    Version:  v5.3.0
    Path:     C:\Program Files\Docker\cli-plugins\docker-compose.exe
  debug: Get a shell into any image or container (Docker Inc.)
    Version:  0.0.47
    Path:     C:\Program Files\Docker\cli-plugins\docker-debug.exe
  desktop: Docker Desktop commands (Docker Inc.)
    Version:  v0.4.1
    Path:     C:\Program Files\Docker\cli-plugins\docker-desktop.exe
  dhi: CLI for managing Docker Hardened Images (Docker Inc.)
    Version:  v0.0.6
    Path:     C:\Users\15L\.docker\cli-plugins\docker-dhi.exe
  extension: Manages Docker extensions (Docker Inc.)
    Version:  v0.2.31
    Path:     C:\Program Files\Docker\cli-plugins\docker-extension.exe
  init: Creates Docker-related starter files for your project (Docker Inc.)
    Version:  v1.4.0
    Path:     C:\Program Files\Docker\cli-plugins\docker-init.exe
  mcp: Docker MCP Plugin (Docker Inc.)
    Version:  v0.43.1
    Path:     C:\Program Files\Docker\cli-plugins\docker-mcp.exe
  model: Docker Model Runner (Docker Inc.)
    Version:  v1.2.5
    Path:     C:\Program Files\Docker\cli-plugins\docker-model.exe
  offload: Docker Offload (Docker Inc.)
    Version:  v0.6.7
    Path:     C:\Program Files\Docker\cli-plugins\docker-offload.exe
  pass: Docker Pass Secrets Manager Plugin (beta) (Docker Inc.)
    Version:  v0.2.0
    Path:     C:\Program Files\Docker\cli-plugins\docker-pass.exe
  sandbox: "docker sandbox" is deprecated, use Docker Sandboxes instead (Docker Inc.)
    Version:  v0.13.0
```

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```
Server:
 Containers: 46
  Running: 0
  Paused: 0
  Stopped: 46
 Images: 19
 Server Version: 29.6.1
 Storage Driver: overlay2
  Backing Filesystem: extfs
  Supports d_type: true
  Using metacopy: false
  Native Overlay Diff: true
  userxattr: false
 Logging Driver: json-file
 Cgroup Driver: cgroupfs
 Cgroup Version: 2
 Plugins:
  Volume: local
  Network: bridge host ipvlan macvlan null overlay
  Log: awslogs fluentd gcplogs gelf journald json-file local splunk syslog
 CDI spec directories:
  /etc/cdi
  /var/run/cdi
 Swarm: inactive
 Runtimes: io.containerd.runc.v2 nvidia runc
 Default Runtime: runc
 Init Binary: docker-init
 containerd version: e53c7c1516c3b2bff98eb76f1f4117477e6f4e66
 runc version: v1.3.6-0-g491b69ba
 init version: de40ad0
 Security Options:
  seccomp
   Profile: builtin
  cgroupns
 Kernel Version: 6.18.33.2-microsoft-standard-WSL2
 Operating System: Docker Desktop
 OSType: linux
 Architecture: x86_64
 CPUs: 20
 Total Memory: 15.49GiB
 Name: docker-desktop
 ID: 8426088b-7a55-4da0-a9e7-301b5465a3fd
 Docker Root Dir: /var/lib/docker
 Debug Mode: false
 HTTP Proxy: http.docker.internal:3128
 HTTPS Proxy: http.docker.internal:3128
 No Proxy: hubproxy.docker.internal
 Labels:
  com.docker.desktop.address=npipe://\\.\pipe\docker_cli
```

```
C:\Users\15L>docker history ubuntu
IMAGE          CREATED       CREATED BY                                      SIZE      COMMENT
de7345b16e94   2 weeks ago   umoci raw add-layer --image /home/buildd/roc…   179B      Add rock control metadata
<missing>      2 weeks ago   umoci config --image /home/buildd/rockcraft-…   0B        Set annotations
<missing>      2 weeks ago   umoci config --image /home/buildd/rockcraft-…   0B        Set labels
<missing>      2 weeks ago   umoci config --image /home/buildd/rockcraft-…   0B        Set default PATH for bare-based rock
<missing>      2 weeks ago   umoci config --image /home/buildd/rockcraft-…   0B        Set default commands
<missing>      2 weeks ago   umoci config --image /home/buildd/rockcraft-…   0B        Set entrypoint
<missing>      2 weeks ago   umoci raw add-layer --image /home/buildd/roc…   100MB
```

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```
C:\Users\15L>docker network inspect bridge
[
    {
        "Name": "bridge",
        "Id": "71e6d7059d7e23b4ccb65c3423f0b12b946a8b8a597f69c9d7e88eb2c2ef5d8d",
        "Created": "2026-07-30T08:27:08.231058927Z",
        "Scope": "local",
        "Driver": "bridge",
        "EnableIPv4": true,
        "EnableIPv6": false,
        "IPAM": {
            "Driver": "default",
            "Options": null,
            "Config": [
                {
                    "Subnet": "172.17.0.0/16",
                    "Gateway": "172.17.0.1"
                }
            ]
        },
        "Internal": false,
        "Attachable": false,
        "Ingress": false,
        "ConfigFrom": {
            "Network": ""
        },
        "ConfigOnly": false,
        "Options": {
            "com.docker.network.bridge.default_bridge": "true",
            "com.docker.network.bridge.enable_icc": "true",
            "com.docker.network.bridge.enable_ip_masquerade": "true",
            "com.docker.network.bridge.host_binding_ipv4": "0.0.0.0",
            "com.docker.network.bridge.name": "docker0",
            "com.docker.network.driver.mtu": "1500"
        },
        "Labels": {},
        "Containers": {},
        "Status": {
            "IPAM": {
                "Subnets": {
                    "172.17.0.0/16": {
                        "IPsInUse": 3,
                        "DynamicIPsAvailable": 65533
                    }
                }
            }
        }
    }
]
```

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Conclusion:

In this experiment, I learnt the fundamentals of Docker by performing basic image and container operations. I verified the Docker installation, searched for and downloaded Docker images from Docker Hub, listed locally available images, and created an Ubuntu container using a Docker image. Inside the container, executed basic Linux commands to navigate directories, create folders and files, write data to a file, and verify its contents. This experiment helped us understand the concepts of Docker images, containers, and containerization, as well as how Docker provides an isolated and consistent environment for running applications efficiently.